Quotient rule

1

a $u' = 3$

b $v' = 4$

c $y' = -\frac{25}{(4x-3)^2}$

d No

2

a $u' = 5$

b $v' = 8$

c $y' = -\frac{45}{(8x-9)^2}$

d No

3

a $u' = 18x$

b $v' = 2$

c $y' = \frac{9x(x+8)}{2(x+4)^2}$

d Yes

4

a $u' = 8x$

b $v' = 4x$

c $y' = \frac{28x}{(2x^2+5)^2}$

d Yes

5

a $y' = -\frac{24}{(4x-3)^2}$

b $y' = -\frac{7}{(3x-4)^2}$

c $y' = \frac{-2\left(4x^2+3\right)}{(4x^2-3)^2}$

d $y' = \frac{-20x^2-24x-5}{(4x^2-1)^2}$

e $y' = -\frac{17}{(7x+3)^2}$

f $y' = \frac{2x-4x^4}{(4x^3+1)^2}$

g $y' = \frac{-15x^2-20x+18}{(6+5x^2)^2}$

h $y' = \frac{-5}{(9\sqrt{x}+8)^2\sqrt{x}}$

6 $y' = \frac{x^2+4x-8}{(x+2)^2}$

7

a $\frac{dy}{dx} = -3x^{-2}$

b $\frac{dy}{dx} = -\frac{3}{x^2}$

c 0

8

a $\frac{dy}{dx} = -\frac{8}{(8x+3)^2}$

b $\frac{dy}{dx} = -\frac{8}{(8x+3)^2}$

9

a $y' = \frac{7}{(3x-4)^2}$

b No

c $x = \frac{4}{3}$

10

a

i $y' = -\dfrac{12}{(x-6)^2}$

ii No

iii $x = 6$

iv Never

v $(-\infty, 6)$ and $(6, \infty)$

b

i $y' = -\dfrac{20}{(3x-4)^2}$

ii No

iii $x = \dfrac{4}{3}$

iv Never

v $\left(-\infty, \dfrac{4}{3}\right)$ and $\left(\dfrac{4}{3}, \infty\right)$

c

i $y' = \dfrac{20}{(5-2x)^2}$

ii No

iii $x = \dfrac{5}{2}$

iv $\left(-\infty, \dfrac{5}{2}\right)$ and $\left(\dfrac{5}{2}, \infty\right)$

v Never

d

i $y' = -\dfrac{12}{(2x-3)^2}$

ii No

iii $x = \dfrac{3}{2}$

iv Never

v $\left(-\infty, \dfrac{3}{2}\right)$ and $\left(\dfrac{3}{2}, \infty\right)$

11

a $\dfrac{dy}{dx} = \dfrac{2\left(4t^2+3\right)^2\left(60t+4t^2-15\right)}{(5+2t)^6}$

b $\dfrac{dy}{dx} = \dfrac{14}{(2+7x)^{\frac{1}{2}}(2-7x)^{\frac{3}{2}}}$

Gradients and tangents

12 $f'(4) = 0$

13 $f'(1) = -2$

14 $f'(0) = \frac{1}{4}$

15

a $f'(2) = 28$

b Increasing

16 $f'(-5) = 7$

17 $\frac{9}{25}$

18 $y = \frac{1}{36}x + \frac{4}{9}$

19 $y = \frac{41}{49}x - \frac{59}{49}$

20 $g'(2) = \frac{6}{121}$

21 $x = 0,\ \frac{2}{3}$

22 $a = 0,\ -6$

23 $k = 3,\ 27$